$x + y = 10$

$x = y$

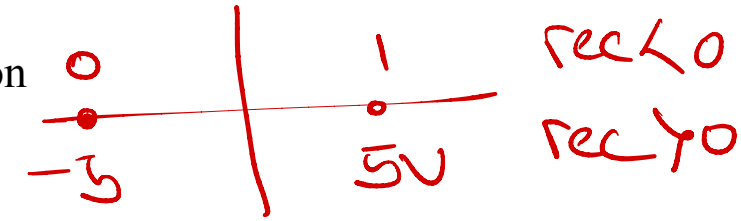
Chapter 1: Experiments, Models & Probabilities

Math 217: Probability & Statistics

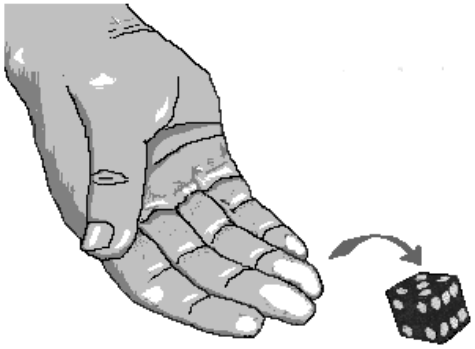
EE/CS Engineering

Machine learning, artificial intelligence, computer graphics, randomized algorithms, image processing, and scientific simulations

- Computer hardware: what is the expected lifetime of a computer circuit composed from many components, each of which has a tiny probability of failure?
- Computer network: what is the probability of a packet being received correctly when being transmitted over an unreliable link? How do we design the transmission protocol so that we recover from transmission errors most efficiently?
- Machine Learning: prediction without using explicit instructions
- In the analysis of communications systems, one is frequently concerned with lower signal-to-noise ratios resulting in "bit-errors", false alarms, etc. There is then a need for careful positioning of thresholds, use of error-correcting codes, etc to insure that the probability of error is kept within acceptable bounds.
  - Depending on threshold value, there is a need to take a decision on which message has been sent to you
  - Examine the PoE values then take a decision



Fundamental Definitions



Random Experiment (Trial): Experiment where the outcomes can not be predicted with certainity. It is called random experiment.

Outcome: An outcome of an experiment is any possible observation of that random experiment

Sample (Outcome) Space: The sample space of an experiment is the set of all possible outcomes of any trial/experiment. It is represented by the symbol S *or* Ω

Event: An event is a set of outcomes of an experiment (a small part of your experiment results=subset of a sample space)

$A_1 = \{1, 2\}$ $A_2 = \{5, 6\}$

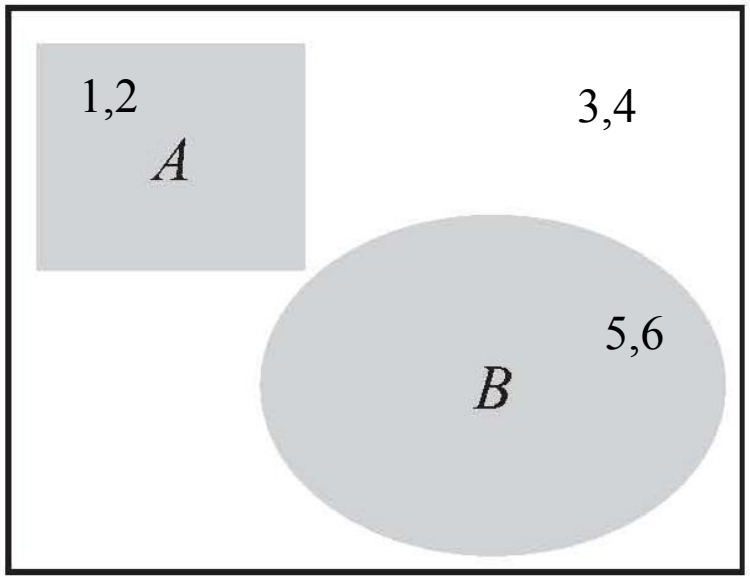
Definition: Mutually Exclusive or Disjoint Events

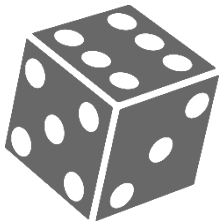
Mutually Exclusive: A collection of event A and B is mutually exclusive event if only if $A \cap B = \emptyset$

S, sample space

Event A is getting a number less than 3

Event B is getting a number greater than 4



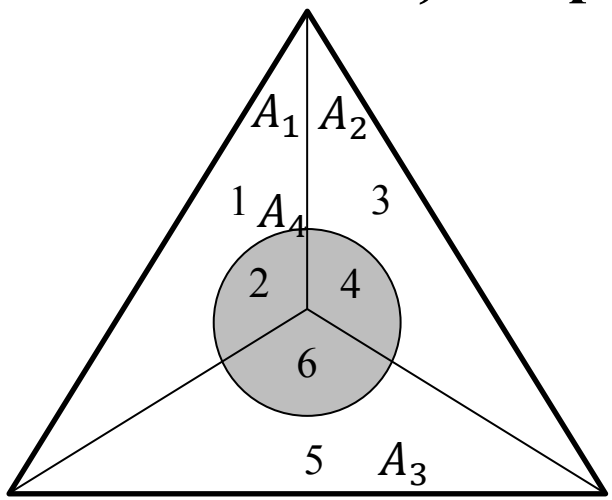


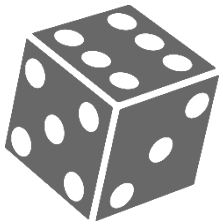
$$A \cap B = \emptyset$$

Definition: Collectively Exhaustive Events

Collectively Exhaustive : A collection of event $A_1, \dots, A_4$ is collectively exhaustive sets if only if $\bigcup_{i=1}^{4} A_i = S$

S, sample space

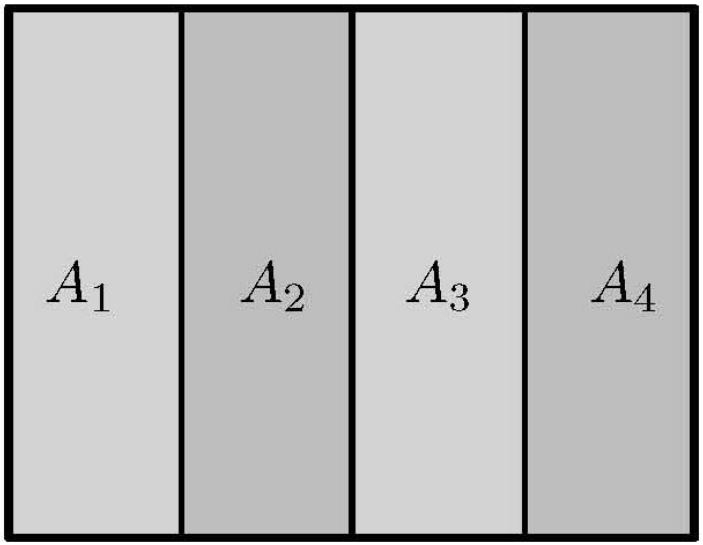




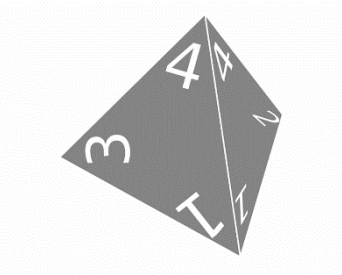
$$A_1 \cup A_2 \cup A_3 \cup A_4 = S$$

Definition: Mutually exclusive and Exhaustive events

Partitions: A collection of event $A_1, \dots, A_4$ is partition if it is both mutually exc. and collectively exh.



S, sample space



$$A_1 \cup A_2 \cup A_3 \cup A_4 = S$$

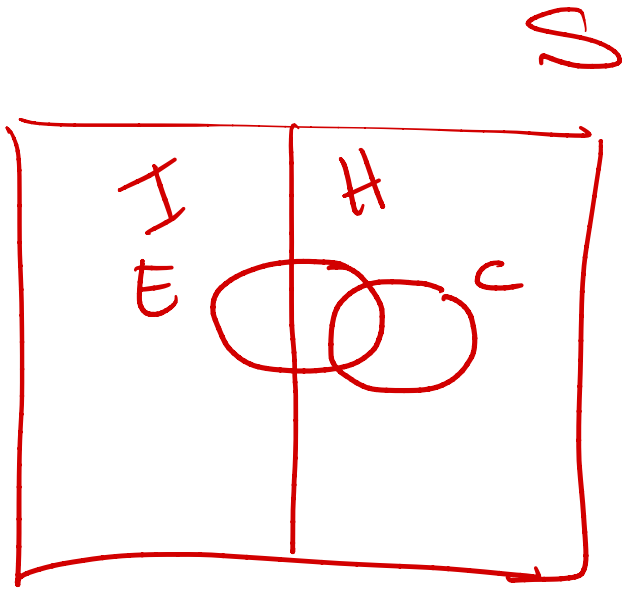
$$A_1 \cap A_2 \cap A_3 \cap A_4 = \varnothing$$

Example-1.1

A smart phone shop offers customers two kinds of phone, iPhone (I) and HTC (H). It is possible to buy

- a iPhone with an optional battery (E)
- a HTC with an optional battery (E) and/or memory card (C)

Draw a Venn diagram to show the relationship among the items (I), (H), (E) and (C).



Experiment-1.1

Observe the phone shop until three customers buy phones (just take brands into consideration). What are the possible outcomes?

$$\text{Outcome } 1 = \{I, I, I\}$$

$$\text{Outcome } 2 = \{I, I, H\}$$

$$\vdots$$

$$\text{Outcome } 8 = \{H, H, H\}$$

Sample space

The sample space of the first experiment is

$$S = \{III, IIH, IHI, IHH, HII, HIH, HHI, HHH\}$$

3 2 2 1 1 1 0

Experiment-1.2

Observe the **number of sold iPhone** until three customers buy phones. What are the possible outcomes?

$$\text{Outcome } 1 = \{0\}$$

$$\text{Outcome } 2 = \{1\}$$

$$\text{Outcome } 3 = \{2\}$$

$$\text{Outcome } 4 = \{3\}$$

Sample space

The sample space of the second experiment is

$$S = \{0, 1, 2, 3\}$$

Probability Axioms

- **Axiom 1:** For a sample event A, $P(A) \geq 0$
- **Axiom 2:** $P(S) = 1$

$0 \leq P(A) \leq 1$

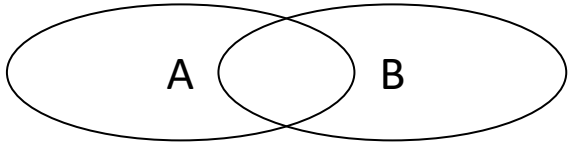
Addition Rule

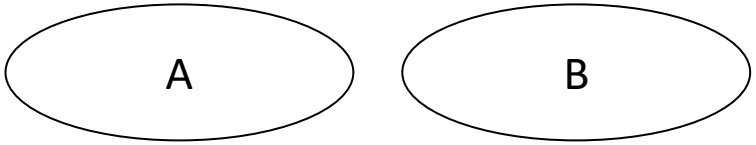
- Probability of a Union

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If two events are mutualy exclusive

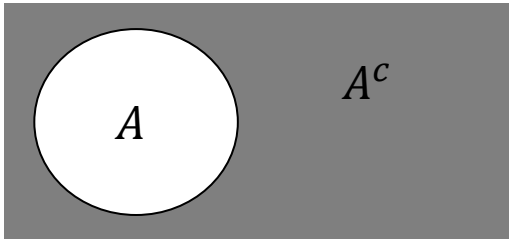
$$P(A \cup B) = P(A) + P(B)$$



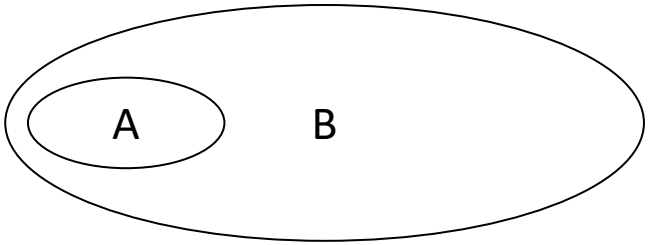


The axioms imply that

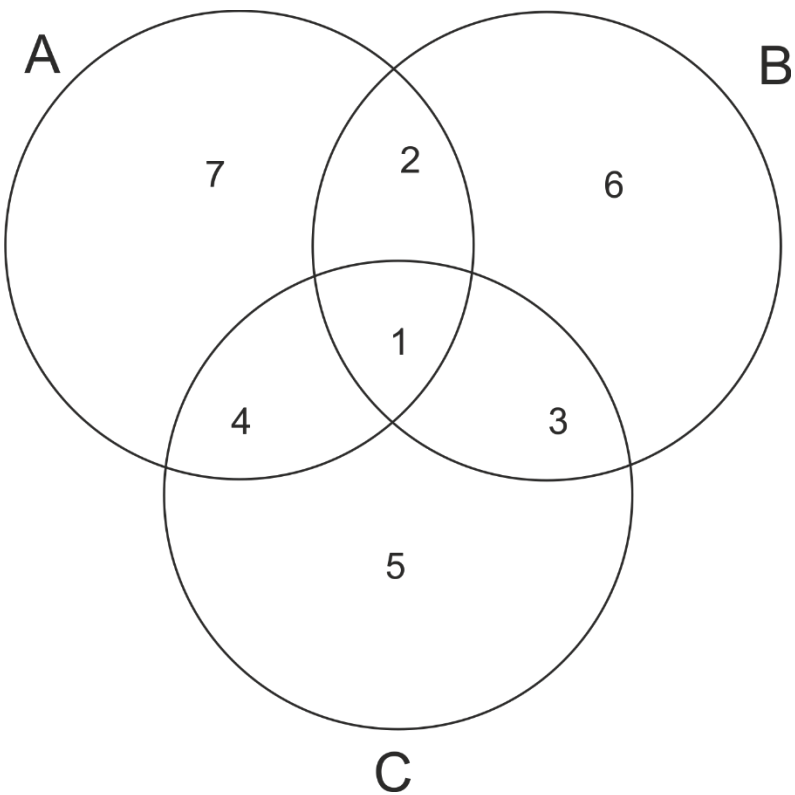
- $P(\emptyset) = 0$
- $P(A^c) = 1 - P(A), \quad P(A)+P(A^c)=1$



- If A is subset of B $P(A) \leq P(B)$



- Let



$A \cup C = \{1, 2, 3, 4, 5, 7\}$

$B^c \cap A = \{7, 4\}$

$A \cap B \cap C = \{1\}$

$(A \cap B) \cap C^c = \{2\}$

$P(A \cup C) = 6/7$

$P(B^c \cap A) = 2/7$

$P(A \cap B \cap C) = 1/7$

$P((A \cap B) \cap C^c) = 1/7$

Instructional Example

Four bits are transmitted over a digital communications channel. Each bit is either distorted or received without distortion. Let Ai denote the event that the ith bit is distorted, i = 1, …, 4.

- Describe the sample space for this experiment.
- Are the Ai's mutually exclusive?
- Describe the outcomes in each of the following events:
- A1 ∩ A2 ∩ A3 ∩ A4 → {d, d, d, d}
- (A1 ∩ A2) ∪ (A3 ∩ A4)

$S = \{$ dddd, ndddd, dddn, nddn, ddnd, ndnd, ddnn, ndnn, dndd, ndd, dndn, ndn, dnnd, nnd, dnnn, nnn $\}$

$P(A_1 \cap A_2 \cap A_3 \cap A_4) = 1/16$

Theorem 1.3

For an experiment with sample space S= $\{s_1, s_2, \dots, s_n\}$ where each outcome s_i is equally likely .

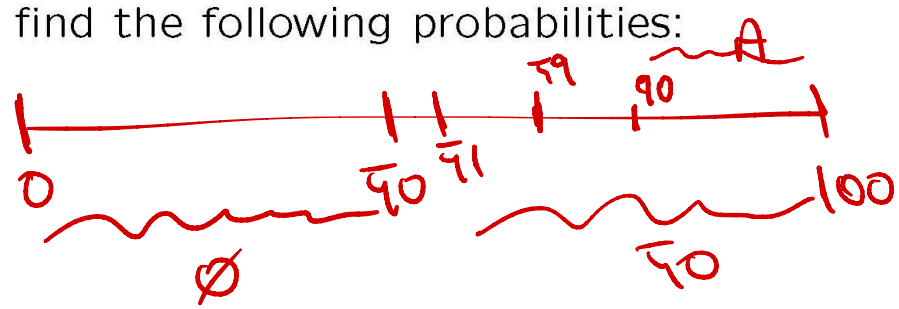
$$P(s_i) = 1/n$$

Example-1.2

Roll a six-sided die in which all faces are equally likely. What is the probability of each outcome? 1/6 Find the probabilities of the events: "Roll 4 or higher," 1/2 "Roll an even number," 1/2

Example-1.3

A student's test score T is an integer between 0 and 100 corresponding to the experimental outcomes $s_0, \ldots, s_{100}$. A score of 90 to 100 is an A, 80 to 89 is a B, 70 to 79 is a C, 60 to 69 is a D, and below 60 is a failing grade of F. If all scores between 51 and 100 are equally likely and a score of 50 or less never occurs, find the following probabilities:



(a) $P[\{s_{100}\}]$ $= 1/50$

(b) $P[A]$ $11/50$

(c) $P[F]$ $= 9/50$

(d) $P[T < 90]$ $= 1 - P(A) = 39/50$

(e) P[a C grade or better] $= 1 - P(F) - P(D) = 31/50$

(f) P[student passes] $1 - P(F) = 41/50$

a) 1/50
b) 11/50
c) 9/50
d) 39/50
e) 31/50
f) 41/50